

Foundry Local 實戰

AI 驅動 Windows Server 2025 維運



Weithenn Wang



About Weithenn



• Weithenn.org

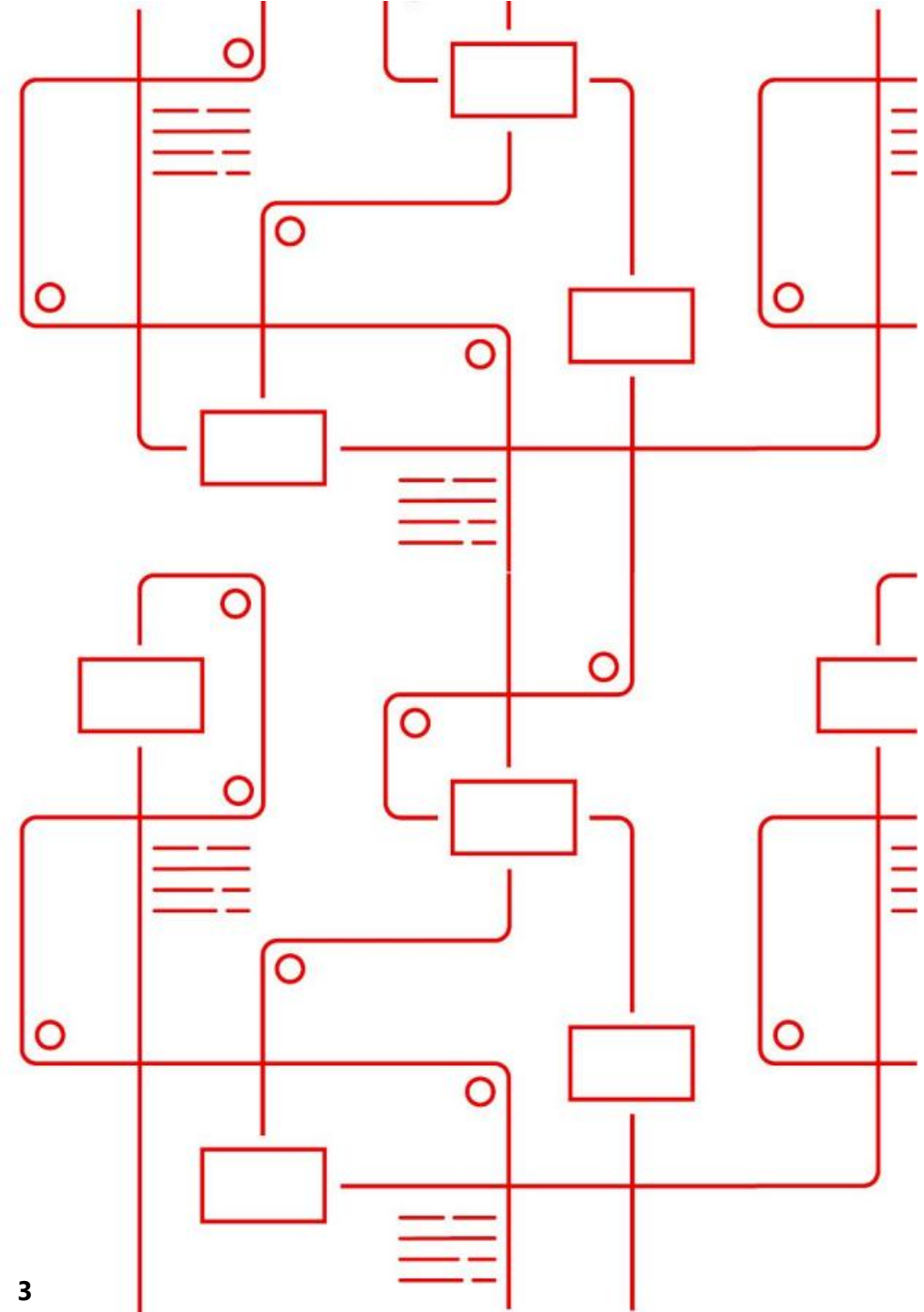
• Microsoft MVP 2012 – 2025

• VMware vExpert 2012 – 2026

• Nutanix NTC 2025 – 2026

• 19 IT books and NetAdmin Magazine

• DevOpsDays Taipei 2026 、 DevOpsDays Tokyo 2026 、 Hello World Dev Conference 2025 、 Kubernetes Summit 2025 、 Cloud Summit 2025 、 DevOpsDays Taipei 2025 、 DevOpsDays Tokyo 2025 、 WebConf Taiwan 2024 、 .NET Conf Taiwan 2024 、 Kubernetes Summit 2024 、 DevOpsDays Taipei 2024 、 SRE Conference 2024 、 DevOpsDays Tokyo 2024.....



What is Foundry Local?

YOUR APPLICATION

Self-contained — ships to end users with no external dependencies

Your Application Code

JavaScript / Python / C# / Rust

REST API (Optional)

OpenAI-compatible HTTP endpoint

Direct function calls (in-process)

Foundry Local Core API

.dll

Windows

.so

Linux

.dylib

macOS

- Single native library — embed directly in your app, no CLI or daemon required
- Manages model lifecycle: load, inference, unload
- Abstracts hardware detection, Execution Provider selection, and caching from your app code
- Thread-safe, session-based API for concurrent inference

Calls into

ONNX Runtime

INFERENCE ENGINE

- Graph partitioning & optimization
- Loads plugin EPs at runtime
- Executes model across best available hardware
- Memory management & data transfer between EPs
- CPU EP always available as fallback

Integrates with

Foundry Catalog

MODEL ACQUISITION

- Cloud-hosted model registry
- Optimized model variants per hardware
- Download on first use, cache locally
- Version-aware — pulls updates when new versions are available

Integrates with

WinML

Windows only

EXECUTION PROVIDER REGISTRATION

- Windows-native ML platform
- Acquires hardware-matched EP plugins from the OS / Windows Update
- Registers EPs with ONNX Runtime at runtime
- Handles driver compatibility & version negotiation

Package

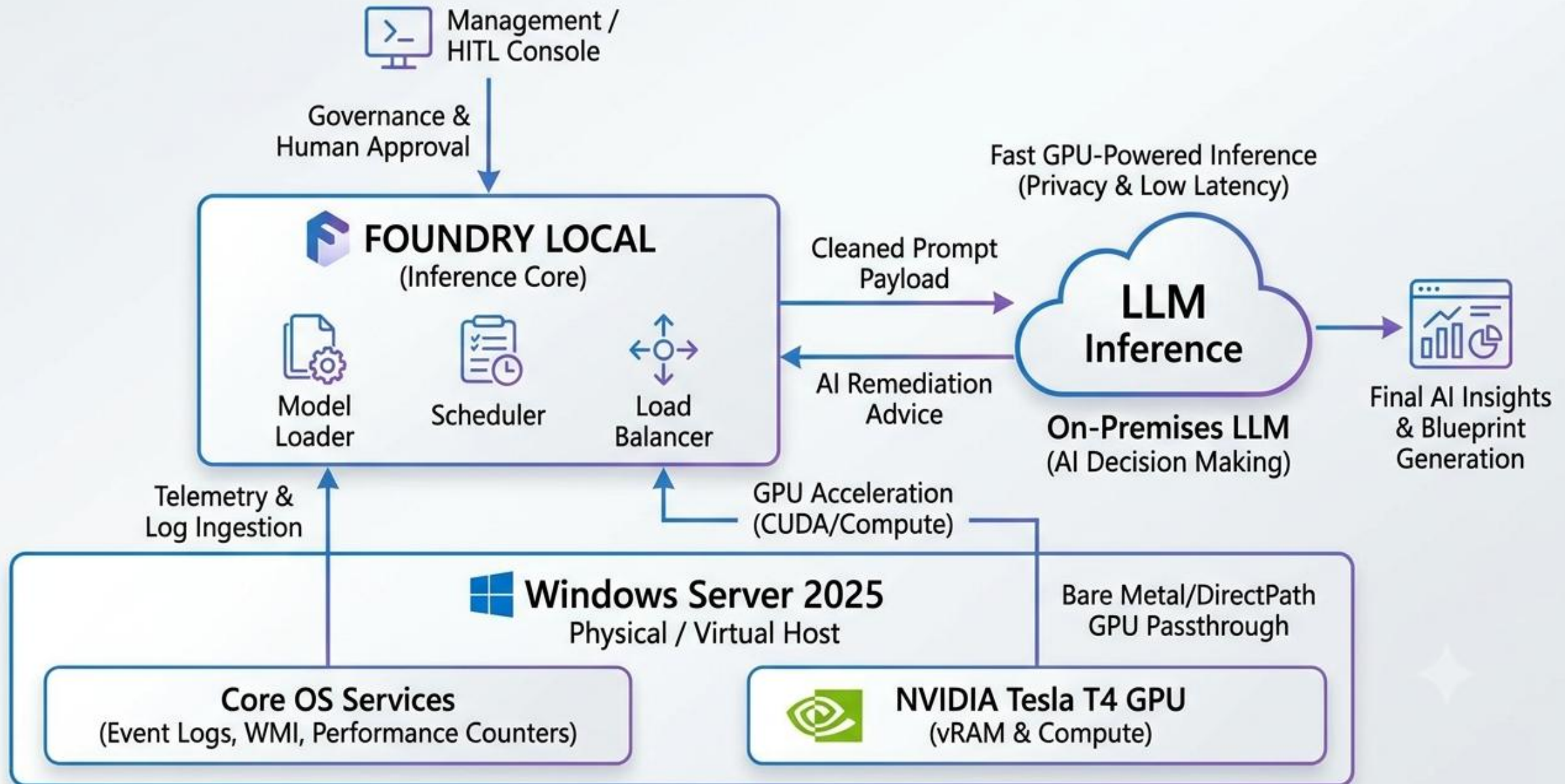
Users



App



On-Premises AI with Windows Server 2025



ON-PREMISES GPU VS. CLOUD GPU: FAULT DIAGNOSIS CONCEPTUAL COMPARISON

ON-PREMISES GPU

(e.g., Tesla T4 on Windows Server 2025)



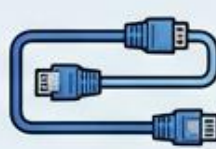
PROS (ADVANTAGES)



HIGH PRIVACY & SECURITY
Telemetry stays on-host.



EXTREME LOW LATENCY
Immediate data access & inference.



OFFLINE CAPABILITY
No reliance on internet connectivity.

CONS (LIMITATIONS)



HIGHER UPFRONT COST
Requires local hardware investment.



MAINTENANCE & UPGRADES
User responsible for physical infrastructure.

CLOUD GPU

(e.g., Azure, AWS)



PROS (ADVANTAGES)



SCALABILITY & ELASTICITY
On-demand resources for diverse workloads.



LOWER UPFRONT COST (OPEX)
Pay-as-you-go model.

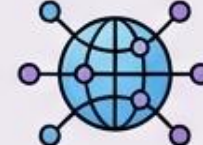


MANAGED INFRASTRUCTURE
Platform handles server maintenance.

CONS (LIMITATIONS)



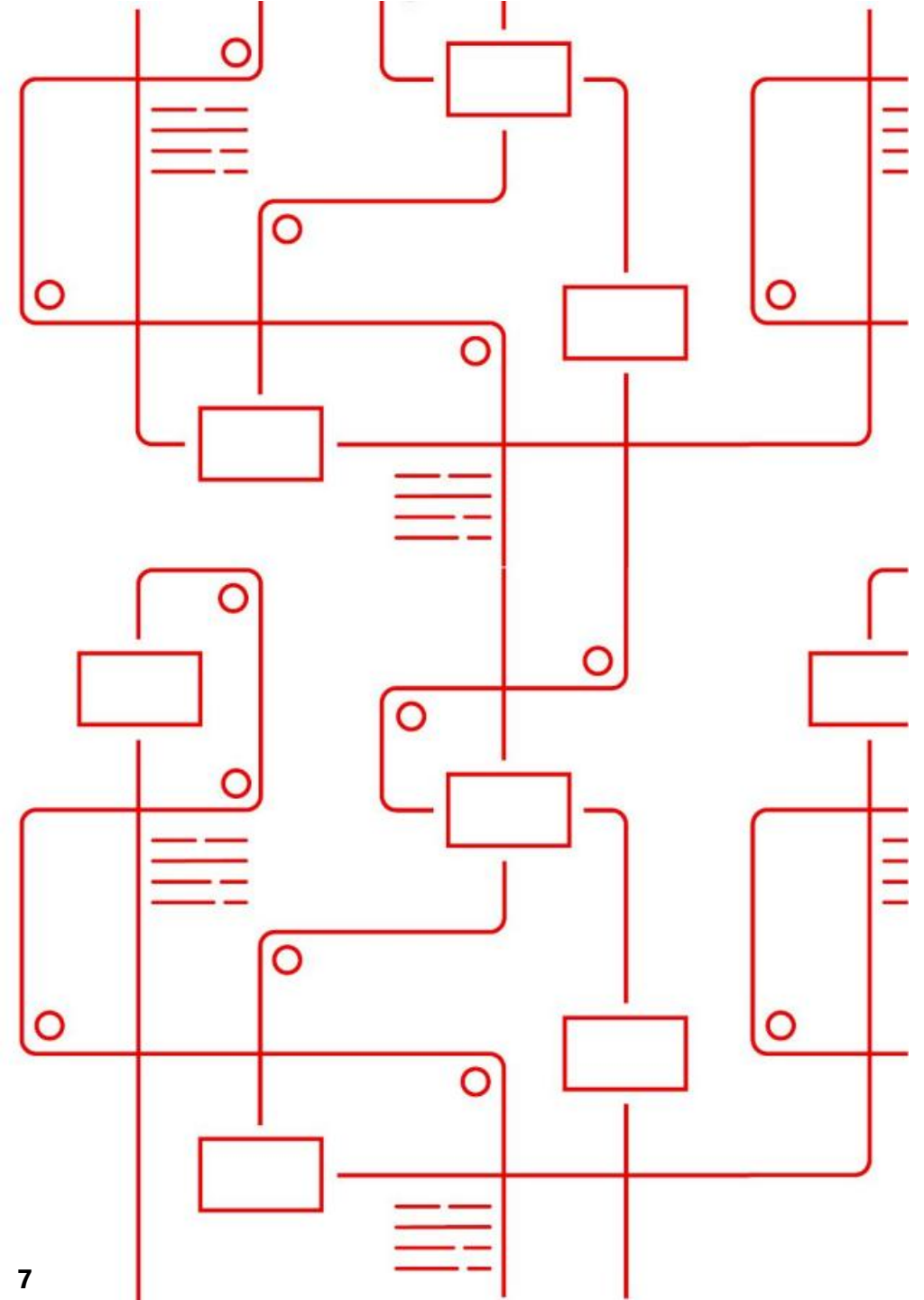
DATA PRIVACY CONCERNS
External data transfer; stricter compliance needed.



NETWORK DEPENDENCY
Requires stable internet connection.



VARIABLE LATENCY
Internet connection affects inference time.



Live Demo

ON-PREMISES IT OPERATIONS: KEY PAIN POINTS.



CRYPTIC LOGS & MANUAL ANALYSIS

ERROR 0x8024200D

0890285088100613592080
048024350000000000000000
0x8024200D0000000000000000
088353000000000000000000
089053000000000000000000
998939500000000000000000
0x8024200D0000000000000000
0x8024200D0000000000000000
049028080000000000000000
60302085081090795335300000
00302085081090795335300000

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- * **DIFFICULT TO DECIPHER:**
Generic Error Codes and Abstract Data.
- * **TIME-CONSUMING:**
Searching for Root Cause
- * **RELIANCE ON EXPERTISE:**
Hard for Juniors to Handle

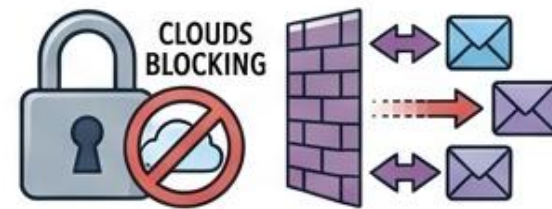


ALERT FATIGUE & OPERATIONAL OVERLOAD



ADMINISTRATOR FALSE POSITIVES

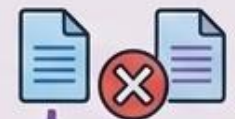
- * **HIGH VOLUME OF ALERTS:**
Distinguishing Threat from Noise
- * **SYSTEM COMPLEXITY:**
Too Many Interfaces to Manage
- * **MANUAL REMEDIATION:**
Risk of Human Error in Fixes.



AIR-GAPPED & COMPLIANCE CONSTRAINTS



SECURE /
OFFLINE ZONE



EXTERNAL DATA
TRANSFER

- * **RESTRICTED INTERNET ACCESS:**
Cannot Leverage Cloud AI Tools.
- * **STRICT DATA PRIVACY:**
Sensitive Logs Must Stay On-Prem.
- * **COMPLIANCE REGULATORY:**
Complex Reporting for Governance

BEHIND THE SCENES: SECURING GPU WARM-UP FOR INSTANT AI RESPONSE



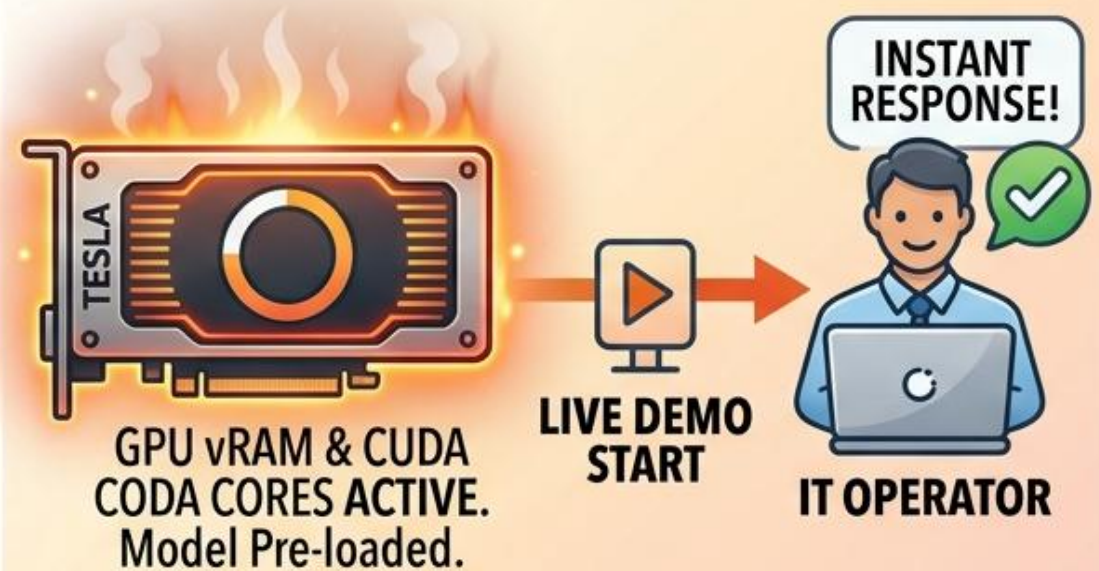
NORMAL LIVE DEMO (No Warm-up)



DELAYED RESPONSE

(Cold Start, vRAM Allocation & Model Loading time)

LIVE DEMO WITH GPU WARM-UP

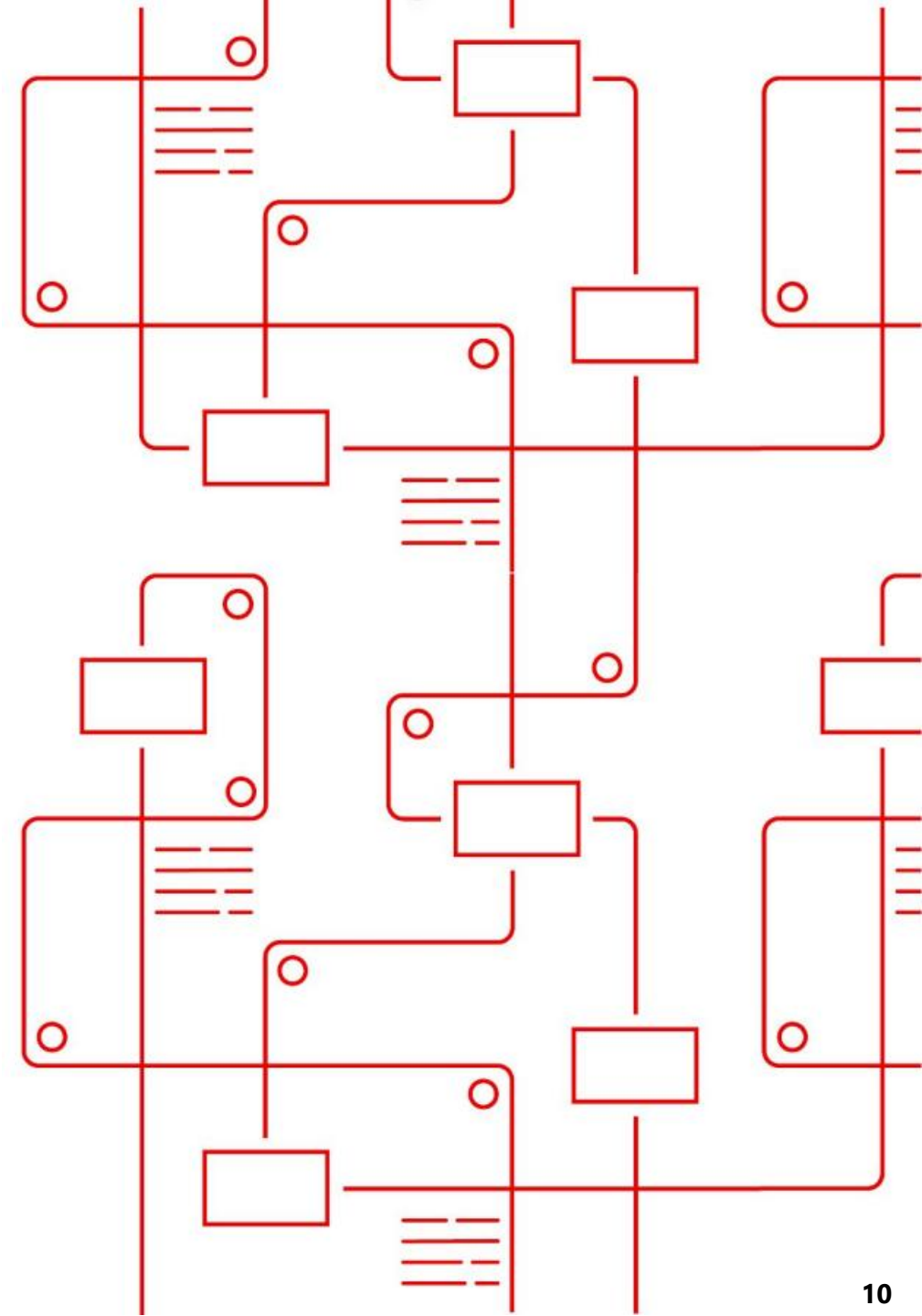


INSTANTANEOUS RESPONSE

(vRAM Ready, Computation starts instantly)

Scenario 1

Local AI Automated Incident Diagnosis



DEMO 1: LOCAL AI FAULT DIAGNOSIS WORKFLOW

SYSTEM ERROR OCCURS



Windows Server 2025
Core Exception
(e.g., Application Error)

EVENT VIEWER LOG GATHERING



[Original Log Payload]
Very cryptic and hard to
read text with hexadecimal
hexadecimal codes and
abstract error
messages.

DIFFICULT
TO READ

LOCAL LLM INFERENCE (via FOUNDRY LOCAL)



FOUNDRY LOCAL



Telemetry cleaned & sent to Local LLM
(Tesla T4 GPU).
Behavioral analysis performed.

[LLM Output]

Clear, readable diagnosis and
step-by-step remediation plan
(e.g., "Restart IIS service and clear
specific cache folder").

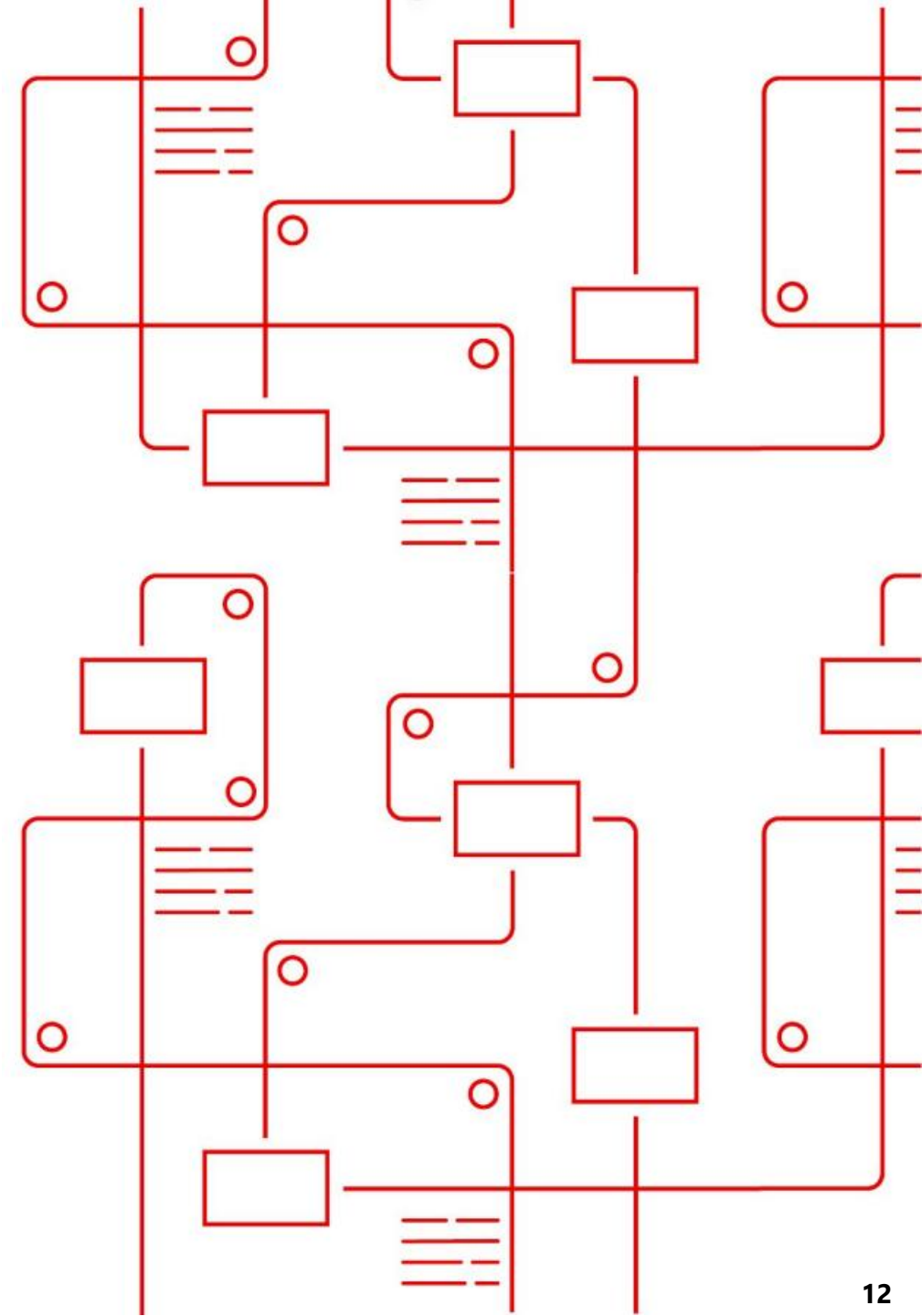


IT OPERATOR EXECUTION

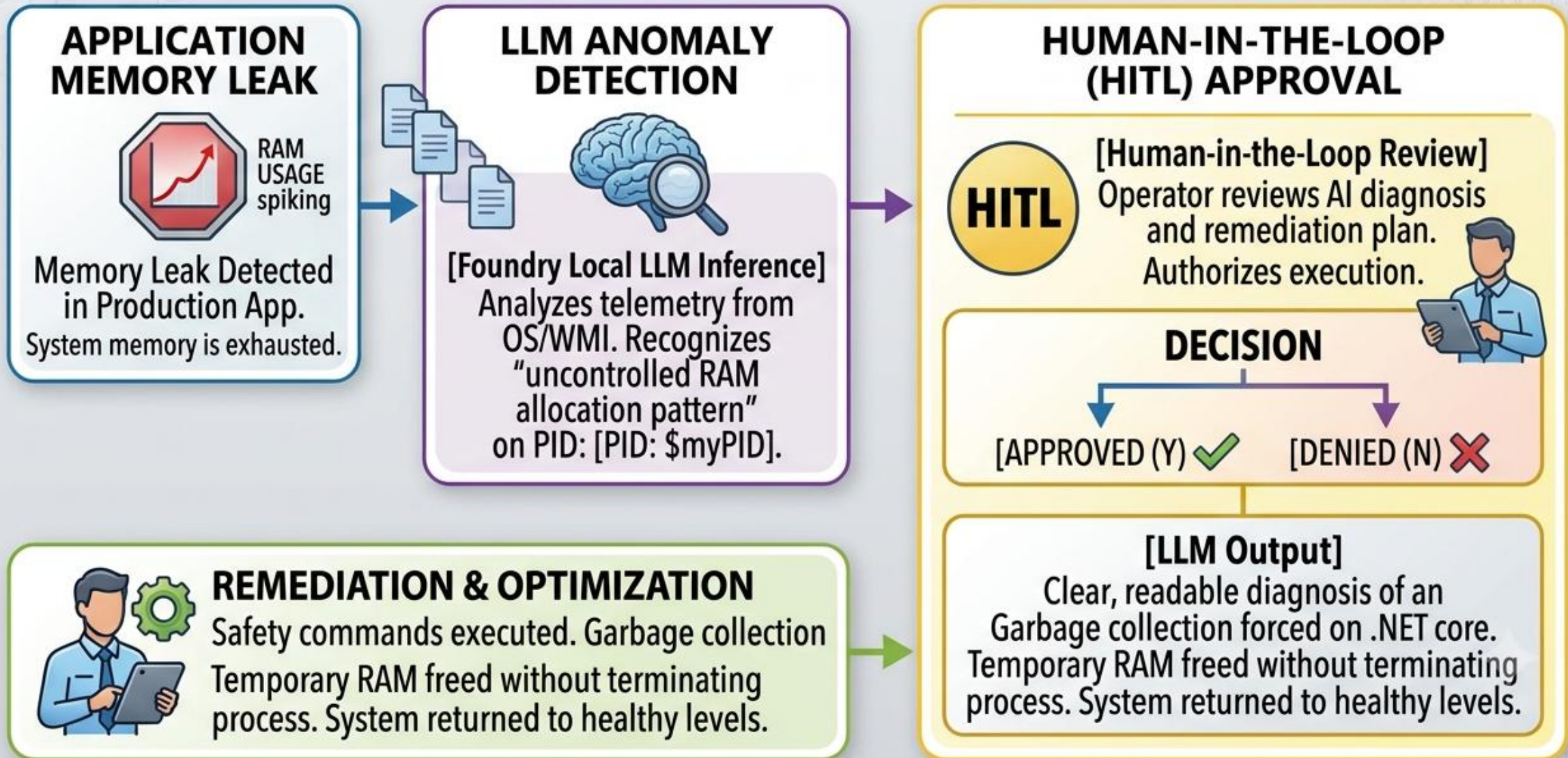
Human-in-the-Loop Review.
IT operator receives clear instructions.
remediation script via WMI/WSS.

Scenario 2

Local AI Proactive Memory Optimization
With Human-in-the-loop (HITL)



DEMO 2: AI PROACTIVE MEMORY OPTIMIZATION WITH HUMAN-IN-THE-LOOP



MEMORY USAGE  WARNING!

MEMORY USAGE

30 GB



 CRITICAL LEVEL!

 USAGE: 30.0 GB

 LIMIT: 32 GB

MEMORY LEAK DETECTED!

MEMORY LEAK DETECTED!



AIops MEMORY LEAK REMEDIATION WITH HUMAN APPROVAL



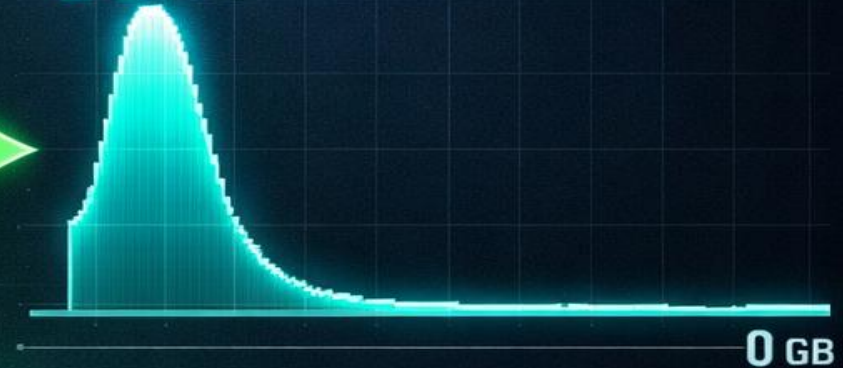
LOCAL LLM - taskkill /PID 9527 /F

PROCESS ANALYSIS



AWAITING CONFIRMATION

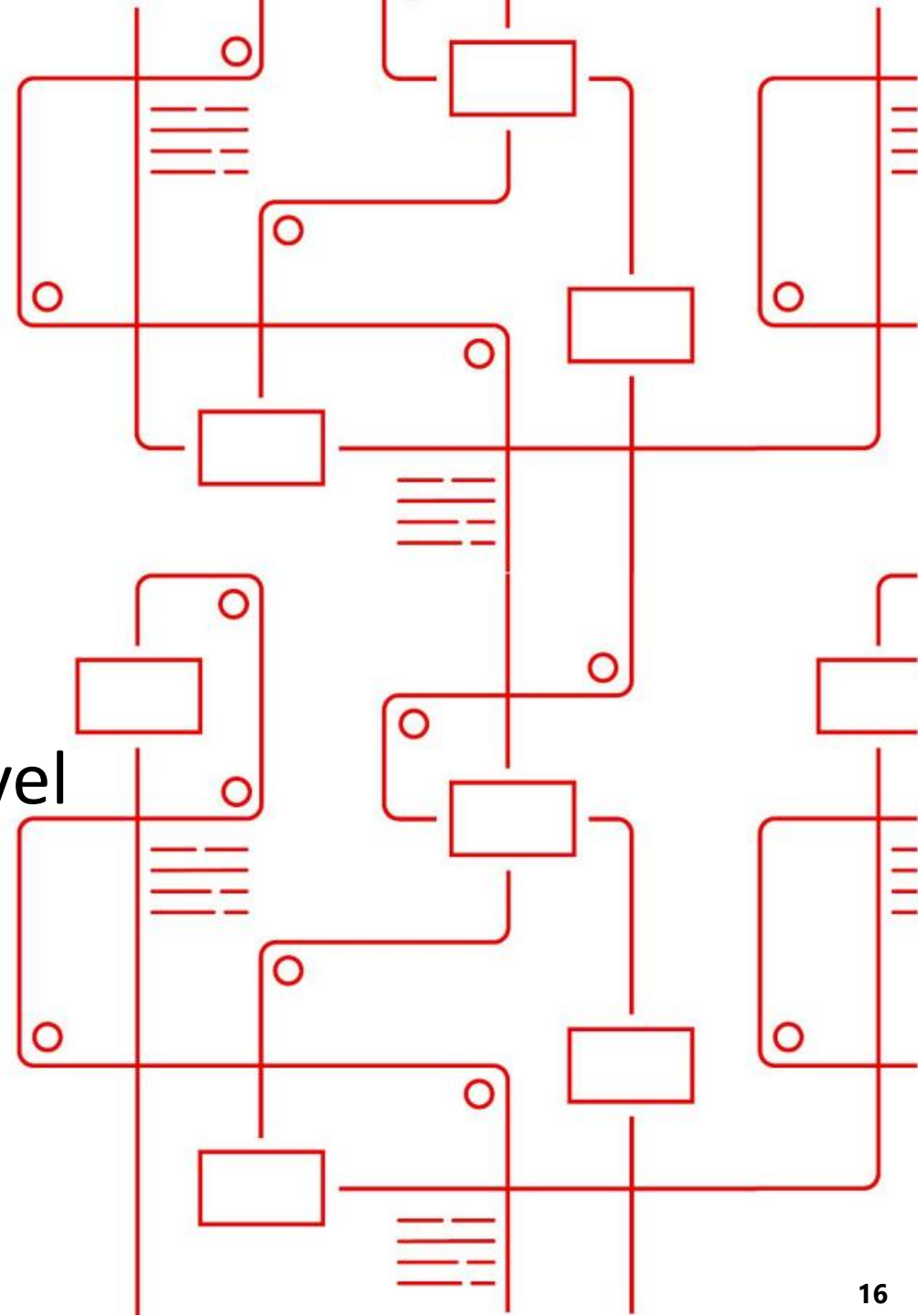
30 GB



MEMORY STABILIZED

Scenario 3

Local AI Security Defense & Blueprint-Level
Auto-Recovery



DEMO 3: LOCAL AI SECURITY DEFENSE & BLUEPRINT-LEVEL AUTO-RECOVERY

MALICIOUS PROCESS THREAT



Obfuscated script attempts to disable host Windows Defender Firewall.

[Threat Action]
Force-disabling all firewall profiles via netsh.

LLM BEHAVIORAL VERDICT



[Foundry Local LLM Inference]
Analyzes telemetry from OS/WMI. Recognizes "unauthorized firewall manipulation pattern".

HUMAN-IN-THE-LOOP (HITL) APPROVAL



[Governance Alert]
Approve AI-recommended remediation commands to force-enable the firewall?



DECISION

[APPROVED (Y) ✓]

[DENIED (N) ✗]

BLUEPRINT-LEVEL AUTO-RECOVERY

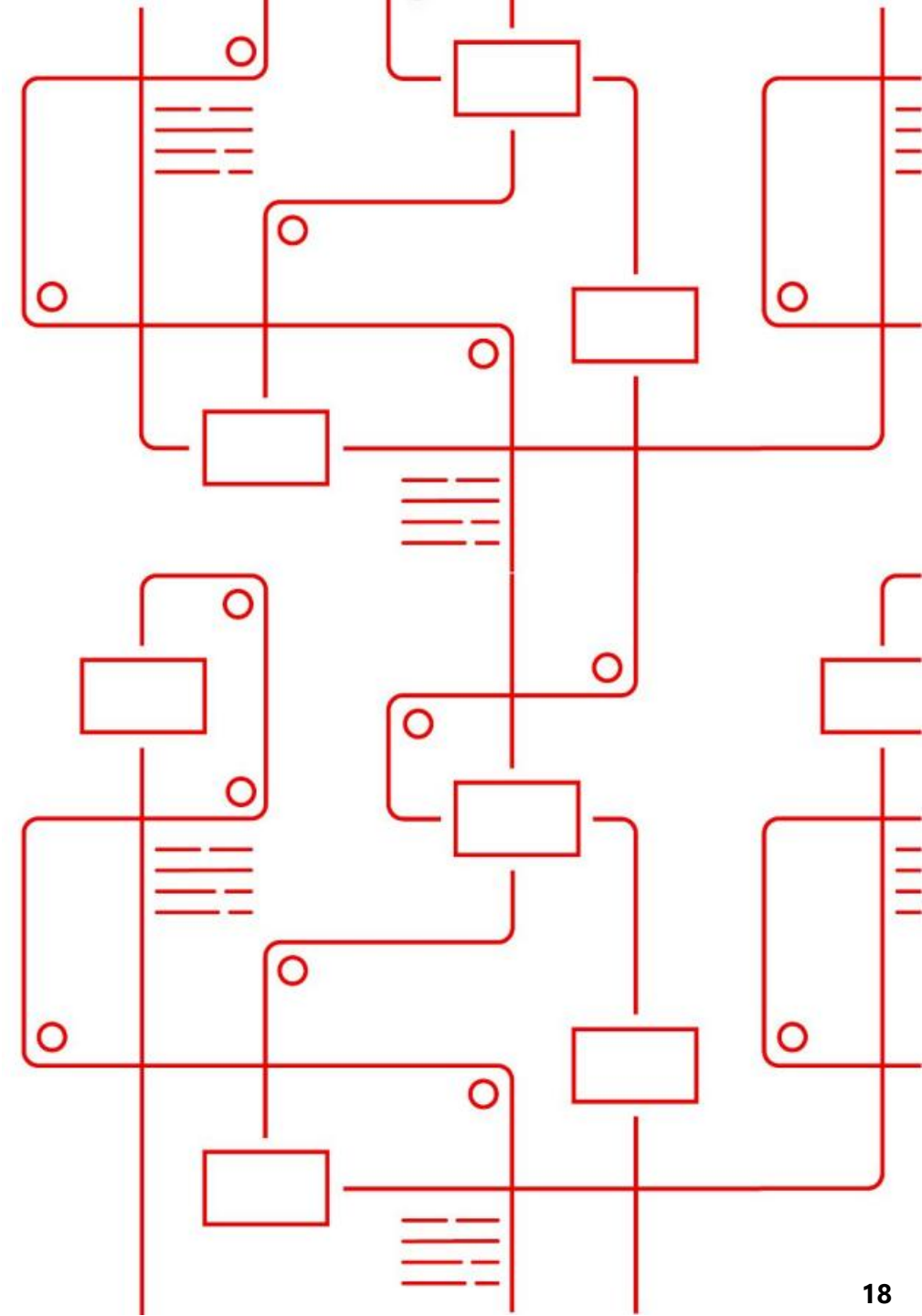


Authorized. Executing host security remediation
[Remediation Success] Defense 100% restored.
(Refresh Firewall Panel: Red turns back to Green!)

[LLM Output]

Clear, readable diagnosis of Windows Firewall profiles restored to secure baseline blueprint! Defense 100% restored.
(Refresh Firewall Panel: Red turns back to Green!)

Resources and More



github.com/Weithenn



Change your avatar

Weithenn

Weithenn

現任 Microsoft MVP, VMware vExpert, Nutanix NTC 及網管人雜誌技術專欄作家，曾擔任 DevOpsDays Tokyo 2025, .NET Conf Taiwan 2024.....等研討會講師，著有微軟 S2D 軟體定義儲存技術實戰.....等共 19 本著作。

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redis-sentinel

Public



Redis Sentinel provides HA (High Availability) for Redis



Shell

☆ 21

🔗 2



powershell

Public



My Buildup or Maintenance Hyper-V, S2D, Azure Stack HCI PowerShell Script

PowerShell

☆ 4

🔗 3



weithenn.github.io

Public



Weithenn Github personal website



HTML

☆ 1



ansible

Public



Weithenn's Ansible Playbooks.

Jinja

☆ 8

🔗 19



EDA-Workshop

Public



EDA(Event-Driven Ansible) Workshop



☆ 5

🔗 5



ollama

Public



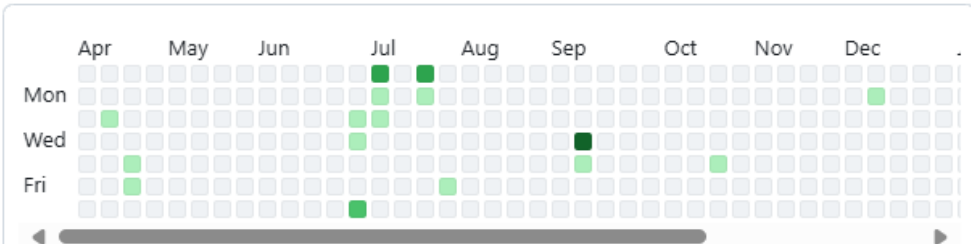
Using docker-compose running ollama and open-webui (COSCUP 2024)

☆ 4

100 contributions in the last year

Contribution settings

2025



2024

2023

2022

Resources

- [What is Foundry Local? - Foundry Local | Microsoft Learn](#)
- [Foundry Local architecture overview - Foundry Local | Microsoft Learn](#)
- [Get started with Foundry Local - Foundry Local | Microsoft Learn](#)
- [Use the Foundry Local CLI \(preview\) - Foundry Local | Microsoft Learn](#)



THANK YOU FOR ATTENDING!

議程：Foundry Local 實戰：AI 驅動 Windows Server 2025 維運



FOUNDRY LOCAL

(Inference Core)

On-Premises LLM & AI Decision Making

- Full OS Integration

[cite: 1, <IMAGE 1, <IMAGE-3>]



Zero-Trust Security

- Local Security Defense & Blueprint-Level Recovery

[cite: 1, <IMAGE-5>, <IMAGE-4>]



Governance & HITL:

Complete Control & Approved Remediation

[cite: 1, <IMAGE-6>, <IMAGE-4>]



Zero-Trust Security

- Local Security Defense & Blueprint-Level Recovery
- 100% On-Host Privacy

[cite: 1, <IMAGE 5>, <IMAGE-4>]



GPU Acceleration

- GPU Warm-up for Instant Response
- Proactive Memory Optimization

[cite: 1, <IMAGE 6>, <IMAGE-2>]

Final Takeaway: Seamless, secure, accelerated IT operations on Windows Server 2025, powered by Foundry Local and On-Premises AI [cite: 1].